

## CSE 342: Computer Networking Lab

### Project Proposal

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| CO  | Description                                                                                                            | PO  | KPA     | Delivery Methods                                                                                  | Assessment Tools                               |
|-----|------------------------------------------------------------------------------------------------------------------------|-----|---------|---------------------------------------------------------------------------------------------------|------------------------------------------------|
| CO1 | Demonstrate a broad knowledge of computer networking concepts, hardware, and terminology through lab-based activities. | PO1 | K3/P4/- | Lectures, active discussion, problem-based exercise, and case study.                              | Lab Project, Demonstration, Viva, Final Report |
| CO2 | Inspect and analyze the operation of a range of networking protocols and devices using simulation or real equipment.   | PO2 | K4/P4/- | Lectures, group discussion, presentation, simulation, and analysis.                               | Lab Project, Demonstration, Viva, Final Report |
| CO3 | Plan, configure, and troubleshoot network systems to meet organizational requirements.                                 | PO2 | K4/P5/- | Group discussion, presentation, logical/mathematical problem solving, and hands-on configuration. | Lab Project, Demonstration, Viva, Final Report |

### Scenario

An enterprise is designing a multi-branch network connecting 6 main routers (R1 to R6) in a topology. Each router connects to an internal local area network (LAN) containing at least 3 host PCs. The entire network must be built using Variable Length Subnet Masking (VLSM) derived from the base IP address given below.

To automate host configuration, a DHCP server is configured in each of the router networks to assign IP addresses dynamically. Centralized services are hosted across the network: at least 2 router networks contain a DNS server, and each DNS server operates alongside at least 3 web servers hosting distinct websites. To enable dynamic communication across all 6 routers, the OSPF routing protocol is configured across the network backbone.

Additionally, a dedicated local architecture uses 2 switches to form 3 separate VLANs, ensuring devices inside each VLAN can only communicate within their own VLAN network. Finally, a border setup uses 2 routers to implement NAT-PAT, translating private IP addresses into public IP addresses when sending packets between them.

### Network Requirements

Base IPv4 block: **172.25.0.0/23**. All subnets must be allocated from this block using VLSM.

| Network                | Requirement          | Notes        |
|------------------------|----------------------|--------------|
| Router R1 LAN          | 65 hosts             | Core LAN     |
| Router R2 LAN          | 25 hosts             | Core LAN     |
| Router R3 LAN          | 15 hosts             | Core LAN     |
| Router R4 LAN          | 10 hosts             | Core LAN     |
| Router R5 LAN          | 10 hosts             | Core LAN     |
| Router R6 LAN          | 5 hosts              | Core LAN     |
| Router-to-router links | 4 addresses per link | 6 links      |
| VLAN 10                | 30 hosts             | VLAN segment |
| VLAN 30                | 25 hosts             | VLAN segment |
| VLAN 50                | 10 hosts             | VLAN segment |

| Network        | Requirement              | Notes                               |
|----------------|--------------------------|-------------------------------------|
| NAT router LAN | 20 hosts                 | Private LAN behind the NAT-PAT edge |
| NAT-PAT link   | 4 addresses              | Link between the 2 edge routers     |
| Server subnets | To be sized by the group | DHCP, DNS and web servers           |

**Notes:** Every host count above is the minimum number of usable host addresses the subnet must provide, with the default gateway included. Subnets must not overlap and must all lie within the base block.

### Question 1: IP Addressing and VLSM Network Plan (Weight: 20% Marks)

Perform VLSM subnetting on the base address 172.25.0.0/23. Subnet the address space to allocate IP blocks for the following:

- 6 router-to-router links, 4 addresses each
- 6 core LANs: R1 (65 hosts), R2 (25 hosts), R3 (15 hosts), R4 (10 hosts), R5 (10 hosts), R6 (5 hosts)
- Server subnets for the DHCP, DNS and web servers
- 3 VLAN segments: VLAN 10 (30 hosts), VLAN 30 (25 hosts), VLAN 50 (10 hosts)
- The NAT router LAN (20 hosts) and the NAT-PAT link (4 addresses)

Prepare a complete IP Addressing Table showing the Network Address, Broadcast Address, Subnet Mask, Usable IP Range and Default Gateway for each network.

CO1 | PO1 | KPA: K3/P4/- | CEP Attributes: P1 (Depth of Knowledge in subnetting concepts), P3 (Depth of Analysis for IP address allocation)

### Question 2: DHCP Server Configuration (Weight: 15% Marks)

Implement DHCP servers across the router networks so that the end-user PCs in each of the 6 core LANs (R1, R2, R3, R4, R5, R6) automatically receive their IP address, subnet mask, default gateway and DNS server address, using the subnets from your Question 1 plan. Demonstrate automatic IP assignment on the host PCs.

CO1 | PO1 | KPA: K3/P4/- | CEP Attributes: P1 (Depth of Knowledge in automated network services), P7 (Interdependence of DHCP with IP subnets and host gateways)

### Question 3: DNS and Web Server Deployment (Weight: 15% Marks)

Deploy at least 2 DNS servers and at least 6 web servers (3 web servers alongside each DNS server). Configure each web server to host a distinct website, and configure the DNS servers to resolve domain names to their respective web server IP addresses. Verify web access from host PCs using domain names in the web browser.

CO2 | PO2 | KPA: K4/P4/- | CEP Attributes: P1 (Depth of Knowledge in application-layer protocols), P7 (Interdependence of DNS domain mapping and HTTP web access)

### Question 4: OSPF Dynamic Routing Configuration (Weight: 20% Marks)

Configure the OSPF routing protocol across all 6 core routers (R1 to R6) to ensure full dynamic reachability across all interconnected router networks. Verify end-to-end communication by performing successful packet tests between different router LANs.

CO2 | PO2 | KPA: K4/P4/- | CEP Attributes: P1 (Depth of Knowledge in dynamic link-state routing), P7 (Interdependence of OSPF routing paths with multi-router connectivity)

### Question 5: VLAN Architecture and Traffic Isolation (Weight: 15% Marks)

Build a VLAN architecture using 2 switches to create the following 3 separate VLAN networks:

| VLAN ID | Hosts Required |
|---------|----------------|
| VLAN 10 | 30 hosts       |
| VLAN 30 | 25 hosts       |

| VLAN ID | Hosts Required |
|---------|----------------|
| VLAN 50 | 10 hosts       |

Configure access ports and trunk links as needed. Verify that devices connected to the same VLAN can communicate with each other, but packets cannot travel across different VLANs at Layer 2.

*CO3 | PO2 | KPA: K4/P5/- | CEP Attributes: P1 (Depth of Knowledge in Layer-2 switching), P2 (Conflicting requirements between network access and Layer-2 traffic separation)*

### **Question 6: NAT-PAT Edge Configuration (Weight: 15% Marks)**

Configure NAT-PAT across 2 edge routers to mask the private internal IP addresses of the NAT router LAN (20 hosts) using public IP addresses during packet transmission. Verify the translation operation using packet simulation.

*CO3 | PO2 | KPA: K4/P5/- | CEP Attributes: P1 (Depth of Knowledge in IP address translation), P7 (Interdependence of private/public IP pools and NAT mapping)*